

verter (DAC), to name a few. These interfaces, along with currents in the near single-digit micro-amperes, make MCUs even more ideal for IoT. Ganesan says, "There is a lot of developer as well as application migration from eight and 16-bit MCUs to 32-bit ones today." He explains, "The chief reason being that typical modern-day or IoT type of applications require more memory (determined by the address bus size), as well as larger data buses, for quicker and more efficient operation from a lay-person's standpoint."

MCUs also offer high-resolution ADC to 24-bit sigma delta ADC, high-resolution DAC, real-time clock (RTC) and liquid crystal display (LCD) controllers. Some companies also offer real-time operating system (OS) free with 32-bit MCUs.

PHY integration reduces board space, power consumption and cost

The value of the IoT comes from its interconnectedness. In many cases, connectivity substantially enhances the utility, serviceability and versatility of embedded systems. An IoT system must often be able to connect in two directions: down to its sources of data such as sensors or other IoT devices, as well as up into the cloud to a data aggregator or centralised control point, informs Joe Folkens, product marketing engineer, Texas Instruments, in his whitepaper. He points out, "Connecting to data sources requires support for a wide range of machine-to-machine (M2M) protocols running over interfaces such as PC, SPI or UART. Connecting to the cloud through the Internet requires an IP based interface, typically Ethernet for wired connectivity and Wi-Fi for wireless applications."

The ubiquity of 10/100 Ethernet makes it a compelling option when secure connectivity to a wired product line is needed. The challenge for developers is to implement this connectivity at the lowest cost and power, without compromising performance

MAJOR CONTRIBUTORS TO THIS ARTICLE

- **Guru Ganesan**, managing director, ARM India
- **Joe Folkens**, product marketing engineer, Texas Instruments
- **Sanjay Gupta**, R&D director, Automotive & Industrial MCU Group, India Design Center, Freescale Semiconductor India Pvt Ltd

or reliability.

Traditionally, the Ethernet MAC has been integrated onto MCUs to reduce cost and design complexity. The Ethernet PHY, on the other hand, has been implemented as a separate component due to analogue integration challenges. Many applications, however, could also benefit from integrating the PHY as well.

An example of MCUs with integrated MAC/PHY is TI's TM4C129x MCU family. It provides an effective solution for connecting systems to the cloud so they can take full advantage of the capabilities enabled by the IoT.

How design engineers are benefited

With onboard or on-chip BLE, ZigBee or Wi-Fi capabilities, and the various interfaces for hooking up sensors, these tiny development boards can be used for both prototyping and production, drastically reducing the time-to-market which the production versions take. Ganesan says, "They help designers to focus more of their energy and time on the real challenges at hand and the systems that can overcome these challenges, than worry about sourcing individual hardware components, manufacturing boards with smaller form factors and soldering." Moreover, he adds, "Prototype systems resulting from such tiny platforms look neat (without any dangling wires or shabby solder joints) from a demonstration as well as marketing standpoint, surely something that would be very attractive to angel investors or venture capitalists."

Gupta believes that it helps a designer to innovate and develop products without thinking about the backend or the support systems. He says,

"These latest functionalities in MCUs allow a designer to check the performance of the battery and exactly know where it lags (if it does). This ensures good-quality battery and also a longer life cycle."

In the Future

"From a system developer standpoint, I would definitely like to be able to see easy availability of inexpensive and micro or mini sensors to grow the Internet of Things purposefully than by the lure of profits," says Ganesan. There are a whole lot of challenges the world has that need solving. The Internet of Things potentially has the answers to solving these challenges quite quickly, easily and efficiently. However, for that purposeful kind of growth of IoT solutions to happen, the kind of sensors and their associated technologies required need to be worked on as well.

For example, besides the inexpensive and easy availability of tiny sensors, technologies such as micro electro mechanical systems (MEMS), which enable embedding these tiny sensors and/or mechanical parts right on the MCU boards, have to develop and evolve as well, shares Ganesan. He says, "More importantly, the world also needs lots of standards in terms of protocols for handling the various kinds of wireless connectivity that will be in use for communication, the various means of data encryption, storage and retrieval mechanisms." He adds, "These will be required for ensuring everybody is on the same page and the various kinds of security gateways that will be required for ensuring privacy are not infringed upon, or data leaks do not happen by accident or design." ●

The author is a senior technical correspondent at EYF